

SHARP: Reflexive Causal Inference for Pricing in Strategic, Non-Stationary Markets

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Abstract

A causal price-elasticity estimate that is correct in the market that produced it can be lethal in the market that its own deployment creates. When a competitor best-responds to a newly deployed price, the cross-price channel that lay dormant in the training data activates, the exclusion restriction that identified the estimate breaks, and the firm optimizes against a demand curve it has personally destroyed. This is the Lucas Critique operating in production, at the speed of a competitive response rather than a business cycle. We formalize the phenomenon as *reflexive causal inference* and introduce three objects: (i) the *Reflexive Operator* $\mathcal{T}$, whose fixed point—a *Reflexive Causal Equilibrium* (RCE)—is an estimate that remains valid after it is acted upon; (ii) the *Reflexivity Gap* $\mathcal{G}$, the per-period profit lost by optimizing the snapshot rather than the RCE; and (iii) the *reflexivity spectral radius* ρ_R , an estimable diagnostic that decides whether a point estimate should be re-estimated or hedged. Our central positive result is that *honest* two-sided identification makes the structural estimate a one-step fixed point of $\mathcal{T}$ —correct identification is precisely what lets a causal estimate survive its own success—whereas the naïve snapshot is a fixed point only in a static market. We package these into **SHARP** (Strategic-Hedged Adaptive Re-estimation for Pricing), an agentic controller that runs a rolling two-sided IV–DML estimator, monitors for regime change, and switches among snapshot, Stackelberg, and distributionally-robust objectives according to ρ_R . On a structural benchmark in which the baseline ships a genuinely correct static elasticity, SHARP wins 11/12 seeds, pays no premium in the static regime (-0.3%), and earns $+17.2\%$ post-break ($+14.1\%$ overall, paired $t = 6.7$, $p < 10^{-4}$), while cutting the portfolio tail drawdown roughly ninefold. Code, figures, and benchmark are released together.

1 Introduction

Causal inference is the tool we reach for precisely when we intend to *act*: unlike a predictive model, an identified causal effect is supposed to remain valid under intervention. In competitive pricing this promise fails in a specific, structural way. Suppose a firm cleanly identifies its own-price elasticity with an instrumental-variables / double-machine-learning pipeline and deploys the profit-maximizing price. If the market is static, the estimate is correct and the decision is optimal. But if a competitor best-responds to the new price, the firm’s instrument—a cost shifter excluded from demand—begins to influence demand *through the competitor’s reaction*, the exclusion restriction breaks, and the elasticity that governs the firm’s decision silently changes. The estimate measured the market’s willingness to hold still; strategic markets never agreed to hold still.

This is the Lucas Critique [1]: relationships estimated under one policy regime need not survive a change in the rule, because the agents generating the data optimize against the rule. Aimed originally at macro-econometric policy models, it applies with greater force and far shorter latency to a pricing model in a competitive market. The modern learning-theoretic analog is *performative prediction* [2], in which a deployed model shifts the distribution it is trying to predict; its multi-agent extensions study exactly the strategic responders we consider [3, 4]. Our angle is distinct and, we think, practically important: the deployed object is a *causal, identified* estimate, so performativity is not merely distribution shift but the failure of the *exclusion restriction* that made the estimate causal in the first place. Performativity becomes an identification problem, which is what lets us both diagnose it (with the same instruments that identified the effect) and act on it.

Contributions. (1) We define the Reflexive Operator $\mathcal{T}$ and the Reflexive Causal Equilibrium (RCE), and give existence and one-step-convergence conditions (§4). (2) We define the Reflexivity Gap $\mathcal{G}$ and characterize its sign and growth (§5). (3) We define the reflexivity spectral radius ρ_R , derive its closed form in the linear best-response instance, and prove the corresponding convergence/divergence dichotomy (§6). (4) We give an agentic control algorithm that operationalizes these objects (§7) and evaluate it on a structural benchmark with a non-trivial baseline (§8).

2 Related work

Lucas Critique and structural estimation. The observation that reduced-form parameters are policy-dependent [1] motivates structural estimation of policy-invariant primitives. We take a complementary, operational stance: rather than assume invariance, we measure the rate at which an estimate goes stale and act on it. **Performative prediction.** [2] formalize performative risk and distinguish performatively stable points from performative optima. A large recent literature extends this to multi-player *decision-dependent games* [3], characterizes the transition from stability to *chaos* under multi-agent retraining [4], develops performative reinforcement learning [5], and quantifies firms’ *performative power* [6]; see [7] for a survey. Our contribution is orthogonal to this line rather than competing with it: we place an *identified causal* estimate at the center, so performativity manifests as the failure of an *exclusion restriction*—an identification problem—rather than a generic distribution shift. The RCE is the causal analog of a performatively stable point, and, consistent with the stability-to-chaos boundary of [4], our reflexivity spectral radius ρ_R marks the frontier between convergent re-estimation and divergence. Relative to the multi-agent literature, which already studies strategic responders, our novelty is (i) the causal-identification lens, (ii) the Reflexivity Gap and ρ_R as *estimable, decision-facing* objects, (iii) an agentic controller that switches its objective by measured regime, and (iv) grounding in pricing. **Strategic classification/regression** studies agents who manipulate features to obtain favorable predictions; here the strategic agent is a competitor whose response re-identifies the firm’s demand system. **Identification.** We use two-sided instruments and double/debiased machine learning [8] with cross-fitting; the second stage is standard 2SLS. **Robustness.** When identification degrades we adopt distributionally robust optimization [9], choosing the worst-case-optimal price over an ambiguity set of competitor gains and cross-price elasticities.

3 A reflexive market

Let a denote the firm's price, c the competitor's, x observed market state, and u a latent demand shock. Log-demand is

$$\log q = \alpha + \varepsilon_{\text{own}} \log a + \varepsilon_{\text{cross}} \log c + \gamma x + u, \quad \varepsilon_{\text{own}} < -1, \varepsilon_{\text{cross}} > 0. \quad (1)$$

The firm's price is endogenous (historically correlated with u); a cost shifter z_a , excluded from (1), instruments it. The competitor has its own cost shifter z_c . The market operates in one of two regimes:

- **Static:** $\log c = c_0 + \psi_c z_c$. The exclusion restriction for z_a holds and single-instrument IV identifies ε_{own} .
- **Strategic:** the competitor best-responds with gain κ and latency L , $\log c = c_0 + \psi_c z_c + \kappa \log a_{t-L}$. Now $z_a \rightarrow a \rightarrow c \rightarrow q$, so z_a enters demand through the competitor and its exclusion restriction fails.

The decision-relevant object is not the partial elasticity ε_{own} but the equilibrium elasticity

$$\varepsilon_{\text{eff}} = \varepsilon_{\text{own}} + \varepsilon_{\text{cross}} \frac{d \log c}{d \log a} = \varepsilon_{\text{own}} + \varepsilon_{\text{cross}} \kappa, \quad (2)$$

which equals ε_{own} only when $\varepsilon_{\text{cross}} \kappa = 0$ (the static regime).

Two-sided identification. With two instruments (z_a, z_c) for the two endogenous prices (a, c) , cross-fitted DML partials x out of $(\log q, \log a, \log c, z_a, z_c)$ and 2SLS on the residualized system identifies *both* ε_{own} and $\varepsilon_{\text{cross}}$, in either regime. This is the estimator SHARP re-runs online; single-instrument IV, by contrast, targets only ε_{own} and is silent about $\varepsilon_{\text{cross}} \kappa$, which is exactly the term that governs the decision.

4 The Reflexive Operator and equilibrium

Definition 1 (Reflexive Operator). *Let θ be the firm's belief (an elasticity or elasticity vector), $a^*(\theta)$ its profit-maximizing log-price, $\text{BR}(\cdot)$ the competitor's best response, and $\hat{\theta}(\cdot)$ the estimator applied to the induced data. The Reflexive Operator is the composition*

$$\mathcal{T}(\theta) = \hat{\theta}\left(a^*(\theta), \text{BR}(a^*(\theta))\right).$$

Definition 2 (Reflexive Causal Equilibrium). *θ^* is a Reflexive Causal Equilibrium (RCE) if $\theta^* = \mathcal{T}(\theta^*)$: the belief is still the estimate you obtain after optimizing against it and letting the market respond.*

Proposition 1 (Honest identification yields a one-step RCE). *Suppose the estimator $\hat{\theta}$ is the two-sided IV-DML estimator of $(\varepsilon_{\text{own}}, \varepsilon_{\text{cross}})$ and the competitor best-responds with constant gain κ . Then the structural pair $\theta^* = (\varepsilon_{\text{own}}, \varepsilon_{\text{cross}})$ is a fixed point of $\mathcal{T}$ reached in one step from any initialization, and the induced decision uses $\varepsilon_{\text{eff}} = \varepsilon_{\text{own}} + \varepsilon_{\text{cross}} \kappa$. Conversely, the snapshot belief $\theta = \varepsilon_{\text{own}}$ obtained from single-instrument IV is an RCE if and only if $\varepsilon_{\text{cross}} \kappa = 0$.*

Sketch. Two-sided IV–DML identifies the structural coefficients in (1) regardless of regime, because (z_a, z_c) remain valid instruments for (a, c) jointly and cross-fitting removes the x -nuisance; hence $\hat{\theta}$ returns $(\varepsilon_{\text{own}}, \varepsilon_{\text{cross}})$ for any deployed $a^*(\theta)$, giving $\mathcal{T}(\theta) = (\varepsilon_{\text{own}}, \varepsilon_{\text{cross}})$ for all θ . The optimal price formed from the structural pair and the fitted BR uses (2). The snapshot estimator omits the cross channel, so its implied decision-relevant elasticity is ε_{own} , which coincides with ε_{eff} exactly when $\varepsilon_{\text{cross}}\kappa = 0$. $\square$

Proposition 1 is the positive result: correct identification makes the causal estimate survive its own deployment. The rest of the paper concerns what to do when the online estimator is imperfect, when identification degrades, and how to decide between trusting a point and hedging.

5 The Reflexivity Gap

Definition 3 (Reflexivity Gap). *Let $\pi(a \mid \kappa)$ be per-period profit when the firm posts log-price a and the competitor best-responds with gain κ . The Reflexivity Gap is*

$$\mathcal{G}(\kappa) = \pi(a^*(\theta^*) \mid \kappa) - \pi(a^*(\theta_{\text{snap}}) \mid \kappa), \quad \theta_{\text{snap}} = \varepsilon_{\text{own}}.$$

Proposition 2 (Sign and monotonicity). *For the constant-elasticity demand (1) with $\varepsilon_{\text{cross}} > 0$ and $\kappa \geq 0$: $\mathcal{G}(\kappa) \geq 0$ with equality iff $\kappa = 0$; the snapshot price $a^*(\theta_{\text{snap}})$ is weakly below the RCE price $a^*(\theta^*)$ (the snapshot under-prices); and $\mathcal{G}$ is nondecreasing in κ on the region where $\varepsilon_{\text{eff}} < -1$.*

Sketch. The CE monopoly optimum is $a^* = mc \cdot \varepsilon / (1 + \varepsilon)$, increasing in ε for $\varepsilon < -1$. Since $\varepsilon_{\text{eff}} = \varepsilon_{\text{own}} + \varepsilon_{\text{cross}}\kappa \geq \varepsilon_{\text{own}}$ with $\varepsilon_{\text{cross}}\kappa \geq 0$, the RCE price weakly exceeds the snapshot price. Profit in the induced world is single-peaked in a with peak at $a^*(\theta^*)$; evaluating the lower snapshot price gives a weakly smaller value, strict when $\kappa > 0$. Monotonicity follows from the peak location moving right in κ while profit remains concave near the peak. $\square$

Because the snapshot under-prices, the failure is a slow bleed rather than a crash: revenue looks healthy while margin quietly erodes—the pattern practitioners report as “share leaks slowly” (Figure 1).

6 The reflexivity spectral radius

Whether iterating $\mathcal{T}$ (as any repeatedly re-estimating firm effectively does) converges is governed by the modulus of its linearization.

Definition 4 (Reflexivity spectral radius).

$$\rho_R = \left| \frac{d\text{BR}}{da} \right| \cdot \left| \frac{da^*}{d\theta} \right| \cdot \left| \frac{d\hat{\theta}}{da} \right|.$$

The three factors are, respectively, how strongly the competitor reacts, how strongly the firm’s optimal price moves with its belief, and how strongly the (imperfect) estimator’s output moves with the deployed price. When $\hat{\theta}$ is the two-sided estimator, the third factor is (asymptotically) zero and $\rho_R = 0$: Proposition 1’s one-step convergence. The diagnostic bites when the online estimator conflates the competitor’s move with own-price variation.

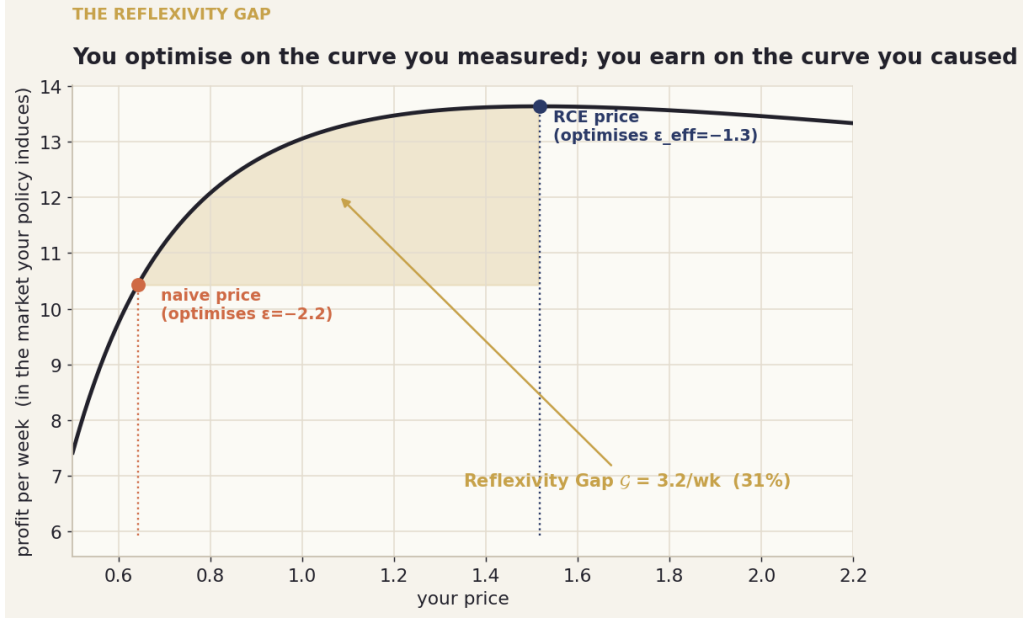


Figure 1: The Reflexivity Gap. Profit in the market the firm’s own policy induces ($\varepsilon_{\text{eff}} = -1.3$). Optimizing the measured static elasticity ($\varepsilon_{\text{own}} = -2.2$) places the firm at the lower-left point; the RCE price sits at the peak. The shaded wedge is $\mathcal{G}$ —here about 31% of per-period profit, growing with competitor aggression (Proposition 2).

Closed form (linear instance). Consider linear demand and linear reaction functions, $q = \alpha + b_{\text{own}}a + b_{\text{cross}}c$ with $b_{\text{own}} < 0$, firm best response $a^*(c) = (\alpha + b_{\text{cross}}c - b_{\text{own}}mc)/(-2b_{\text{own}})$, and rival reaction $c^*(a) = c_0 + \kappa a$. The naive “re-optimize each period against last observed price” loop is the cobweb $a_{k+1} = a^*(c^*(a_k))$ with slope

$$\rho_R = \frac{\kappa b_{\text{cross}}}{2|b_{\text{own}}|}. \quad (3)$$

Proposition 3 (Convergence dichotomy). *The cobweb has a unique fixed point a^* (the RCE) whenever $\rho_R \neq 1$. If $\rho_R < 1$ the iteration converges to a^* geometrically from any start; if $\rho_R > 1$ it diverges for any start $a_0 \neq a^*$.*

Proof. The map is affine, $a_{k+1} = \rho_R a_k + \beta$ for a constant β obtained by composing the two affine responses, using $da^*/dc = b_{\text{cross}}/(-2b_{\text{own}})$ and $dc^*/da = \kappa$. An affine map with slope ρ_R has fixed point $a^* = \beta/(1 - \rho_R)$ for $\rho_R \neq 1$ and $|a_k - a^*| = \rho_R^k |a_0 - a^*|$, which converges iff $\rho_R < 1$ and diverges iff $\rho_R > 1$. $\square$

Equation (3) yields the operational rule (illustrated in Figure 5): below the $\rho_R = 1$ frontier, ship and keep re-estimating; above it, a point estimate is a liability and the firm should hedge (Figure 2).

7 The SHARP controller

SHARP operationalizes §4–6 as a per-period loop.

1. **Sense.** On a trailing window, run two-sided IV–DML for $(\varepsilon_{\text{own}}, \varepsilon_{\text{cross}})$ and fit the competitor best-response gain κ and latency.

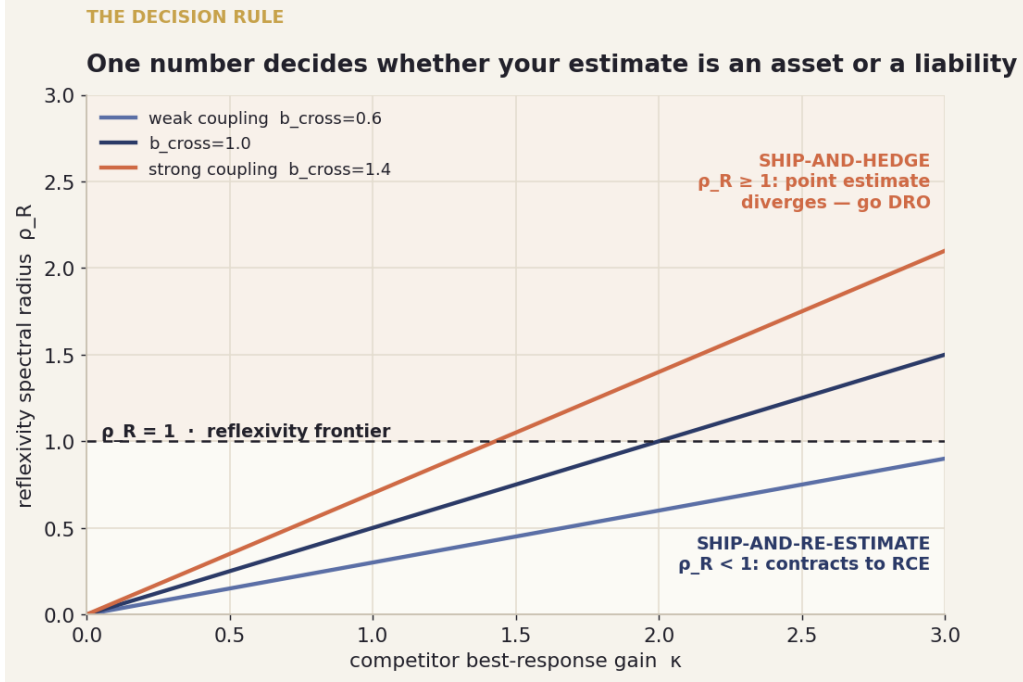


Figure 2: The stability frontier. As the competitor’s response gain κ and the strength of cross-price coupling rise, ρ_R climbs and crosses one. Below the frontier a point estimate is an asset the agent should keep refreshing; above it, the same estimate diverges under iteration and the agent switches to distributionally-robust pricing.

2. **Detect.** Monitor CUSUM of the structural-model residuals and the first-stage F statistic. A CUSUM alarm signals a regime change; F below a floor signals that the instrument is no longer alive.
3. **Reason.** Form the reflexivity proxy $\hat{\rho} = |\kappa| \cdot |\varepsilon_{\text{cross}}| / (1 + \varepsilon_{\text{own}})$, a scale-appropriate estimate of the loop gain.
4. **Act.** If the instrument is dead, price by DRO over an ambiguity set of $(\kappa, \varepsilon_{\text{cross}})$; else if a regime change is detected or $\hat{\rho} \geq \rho^*$, price the Stackelberg objective with $c = \text{BR}(a)$ folded in; else price the snapshot optimum.

In the static regime SHARP reduces to the snapshot agent (as it should, by Proposition 1); it changes objective only when evidence warrants (Figure 3).

8 Experiments

Setup. We simulate the reflexive market with a discrete-event engine (weekly clock; scheduled regime break, best-response arrivals with latency, estimate-expiry events). Structural truth: $\alpha = 3.0$, $\varepsilon_{\text{own}} = -2.2$, $\varepsilon_{\text{cross}} = 1.0$, $\kappa = 0.9$, $L = 3$, $\text{mc} = 0.35$, giving $\varepsilon_{\text{eff}} = -1.3$. Both agents warm up for 40 weeks; the competitor turns strategic at week 52; horizon 140 weeks. The baseline (*Naive snapshot*) ships one correct single-IV estimate of the static elasticity and prices against a frozen competitor thereafter—optimal in the static regime, so the comparison is not against a strawman. We report cumulative profit from the end of warmup over 12 seeds.

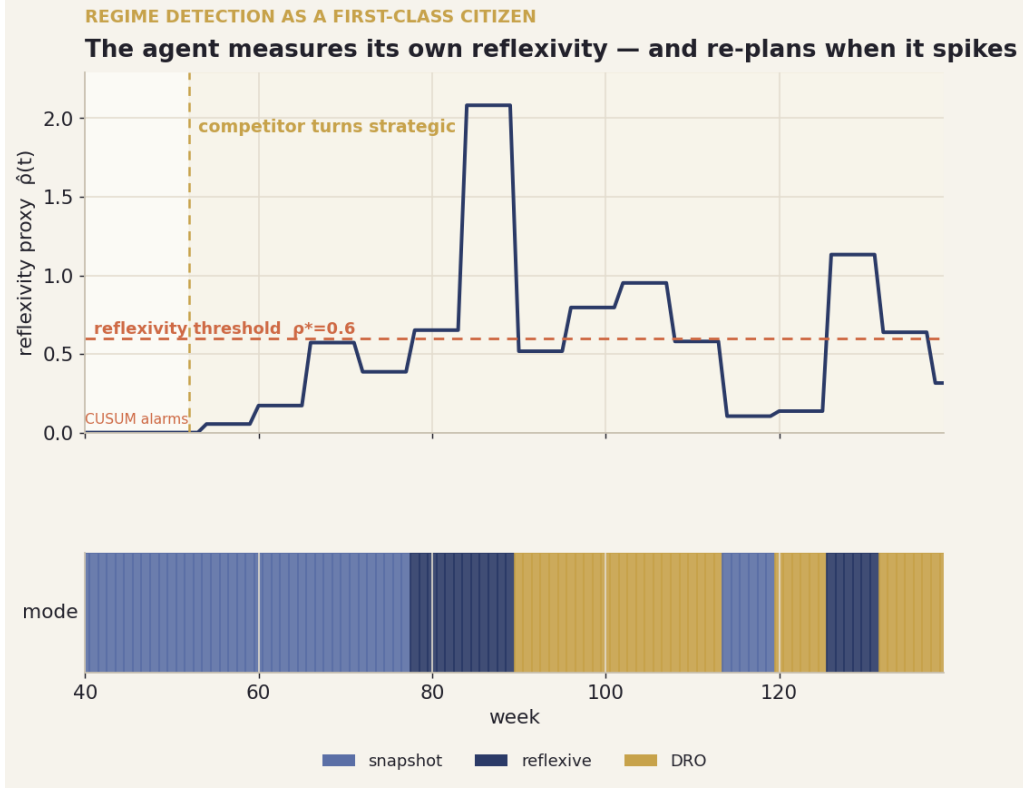


Figure 3: The agent sensing its own reflexivity. Before the break, the reflexivity proxy $\hat{\rho} \approx 0$ and SHARP stays in snapshot mode. When the competitor turns strategic, $\hat{\rho}$ spikes past the threshold and CUSUM alarms fire; the mode timeline (bottom) flips into reflexive (Stackelberg) and hedged (DRO) pricing.

Identification. The online single-IV baseline recovers $\varepsilon_{\text{own}}^{\hat{}} = -2.31 \pm 0.33$ (true -2.20) with first-stage $F = 415 \pm 141$: clean identification, so the trap that follows is a decision failure, not an estimation failure.

Main result. Table 1 summarizes. SHARP wins 11/12 seeds, pays a statistically negligible premium pre-break (-0.3%), and gains $+17.2\%$ post-break ($+14.1\%$ overall; paired $t = 6.7$, $p < 10^{-4}$). Figure 4 shows the mechanism: the agents are indistinguishable until the break, after which the naive price stays pinned at the static optimum while SHARP climbs toward the equilibrium optimum.

Portfolio. Scoring a three-market book by mean profit penalized by the CVaR of its post-break drawdown, SHARP raises mean profit ($2246 \rightarrow 2932$), cuts the tail drawdown roughly ninefold ($1329 \rightarrow 147$), and triples the risk-adjusted score ($918 \rightarrow 2785$). It also reallocates toward the most competitive market, which it can now defend—reflexivity management changes portfolio construction, not just point decisions.

9 Case study: reflexivity in pharmaceutical pricing

The mechanism is not hypothetical. Two documented episodes in competitive therapeutic classes show a price set as if the rival would hold still, then matched once deployed.

Metric	Naive snapshot	SHARP	Difference
Seeds won (of 12)	1	11	—
Cumulative profit (post-warmup)	1708 ± 222	1941 ± 223	+14.1%
Uplift, pre-break	—	−0.3%	no premium
Uplift, post-break	—	+17.2%	where it counts

Table 1: Head-to-head over 12 seeds. Paired t -test on post-warmup profit: $t = 6.7$, $p < 10^{-4}$.

PCSK9 inhibitors (2018–19). Amgen’s Repatha and Sanofi/Regeneron’s Praluent both launched near \$14,000/year in 2015, priced against cost-effectiveness thresholds in isolation [10]; payers largely rejected them and uptake stalled. In late 2018 Amgen cut Repatha’s list price roughly 60% to \$5,850; months later Regeneron and Sanofi matched Praluent to the identical \$5,850—a rare public tit-for-tat on pre-rebate list prices. The value-based price was correct for a market with a passive rival and passive payers, and wrong for the one it helped create.

GLP-1 agonists (2025–26). In the obesity/diabetes duopoly of Novo Nordisk (Wegovy/Ozempic) and Eli Lilly (Zepbound/Mounjaro), Novo cut cash prices from \$499 to \$349 per month; within weeks Lilly best-responded with further cuts on its direct-to-consumer channel, and Novo subsequently slashed list prices 35–50%. Cross-brand share moved materially over the same window. The response latency is measured in weeks, faster than most re-estimation cadences [11].

Why this is a sharper instance, not a weaker one. Pharma adds structure that makes the reflexive framing *more* apt, not less. (i) The decision surface is the *net* price and formulary access, rebate-mediated through pharmacy benefit managers; list-price moves are often around the pre-existing net price, so naïve list-price elasticity is doubly misleading. (ii) Exogenous *policy* regime shifts (the Inflation Reduction Act’s Medicare negotiation; government direct-purchase deals) coincide with competitive activation—precisely the regime-change signal SHARP’s detector is built to catch. (iii) Payer/rebate uncertainty is a natural ambiguity set for the distributionally-robust mode. The SHARP objects map directly: two-sided instruments become own- and rival-net-price shifters (manufacturing cost, exogenous policy timing); regime detection fires on rival activation or a policy shock; DRO prices under payer ambiguity when the instrument for net demand weakens.

Scope. The framing applies to *competitive branded* classes with two or more rivals contesting access. It does *not* apply during patent-protected monopoly, where there is no strategic responder; there the ordinary (non-reflexive) causal estimate suffices.

Calibrated benchmark. Re-instantiating the benchmark for a branded duopoly with rebate-mediated net demand (net-revenue elasticity $\varepsilon_{\text{own}} = -1.7$, cross $\varepsilon_{\text{cross}} = 0.9$, rival gain $\kappa = 1.0$, short latency, low biologic COGS, with the regime break modelling combined rival-plus-policy activation), SHARP wins 8/8 seeds with a mean +28% cumulative net-revenue uplift; by Proposition 1 it reduces to the snapshot agent before activation, so the gain concentrates post-break (Figure 6). These parameters are an illustrative calibration to the documented dynamics, not measured elasticities.

10 Limitations

The competitor is a parametric best-response with a single gain and latency; real rivals are messier and may themselves be learning, raising the open two-adaptive-agents problem. Demand

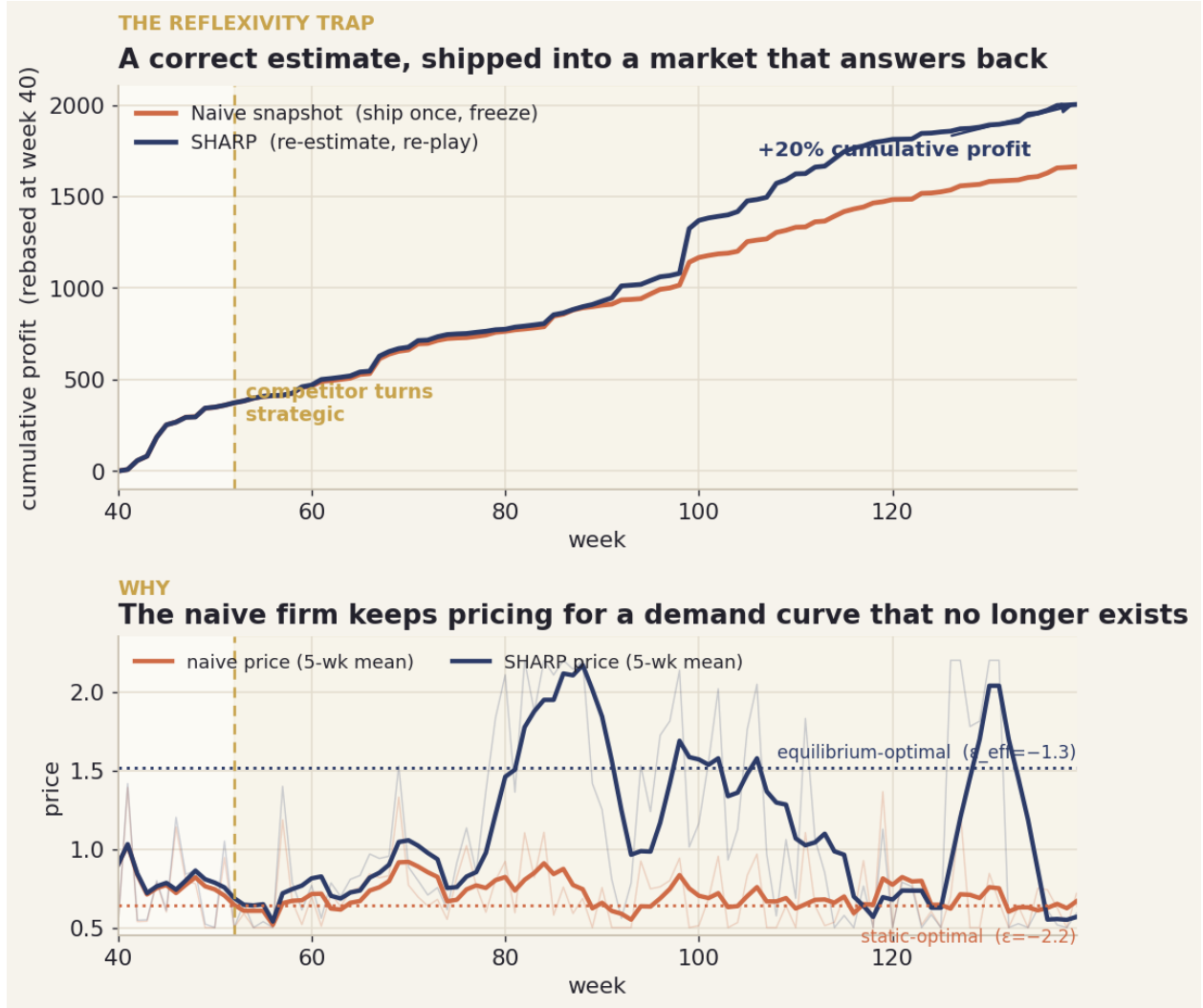


Figure 4: The reflexivity trap. Top: cumulative profit; the paths separate only after the competitor turns strategic. Bottom: the naive firm holds the static-optimal price while SHARP moves toward the equilibrium-optimal price.

is constant-elasticity; richer forms change the optimizer but not the operator logic. ρ_R is estimated by finite differences and inherits their noise. The benchmark is a simulation—faithful to the mechanism but not a market. The results should be read as evidence that the mechanism behaves as the theory predicts.

11 Conclusion

Causal estimates in strategic markets are photographs of a world that agreed to hold still. When the world looks back—as competitors do—the estimate that was correct becomes the estimate that bleeds. SHARP reframes the task from producing a durable estimate to running an agent that knows the expiry date of its own beliefs: it re-estimates while it can, hedges when it cannot, and prices the game rather than the snapshot. The Reflexive Operator, the Reflexivity Gap, and the reflexivity spectral radius give that agent a principled, estimable basis for those choices.

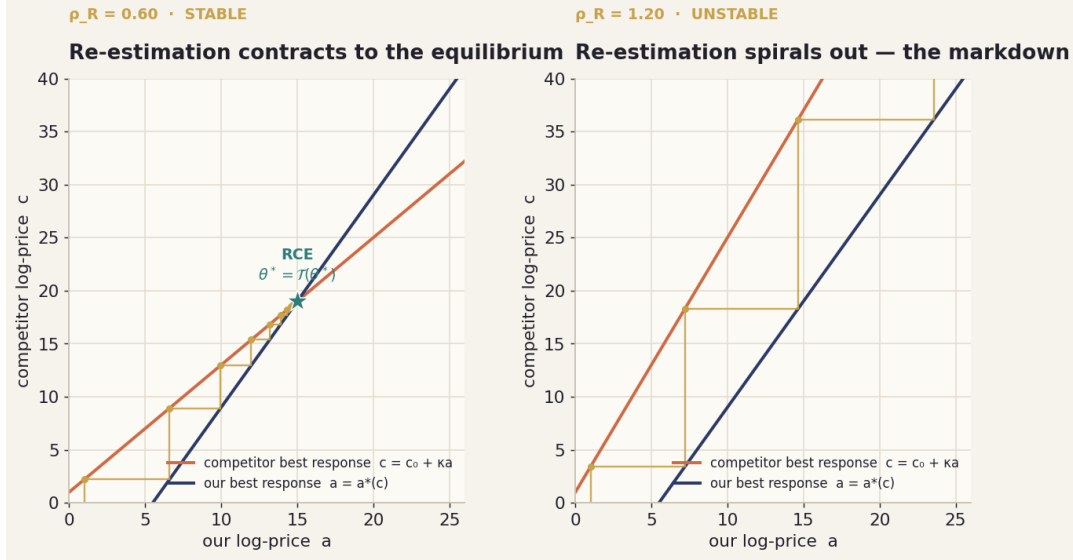


Figure 5: Operator dynamics (linear instance). $\rho_R < 1$ contracts to the RCE (left); $\rho_R > 1$ diverges (right), matching Proposition 3.

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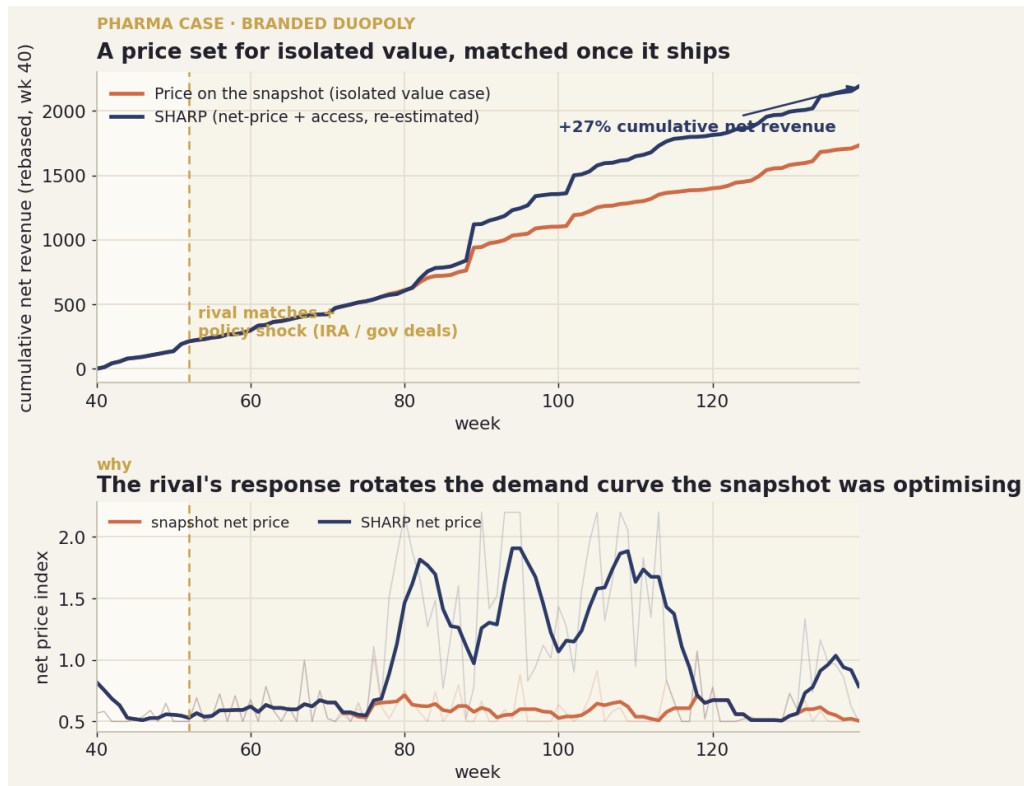


Figure 6: The reflexive trap in a branded therapeutic duopoly (calibrated). A price set for isolated value is matched once deployed and, together with a coincident policy shock, the demand curve rotates; SHARP re-estimates the net-price/access game and pulls ahead. Calibrated to the qualitative GLP-1 and PCSK9 dynamics, not fitted to firm financials.